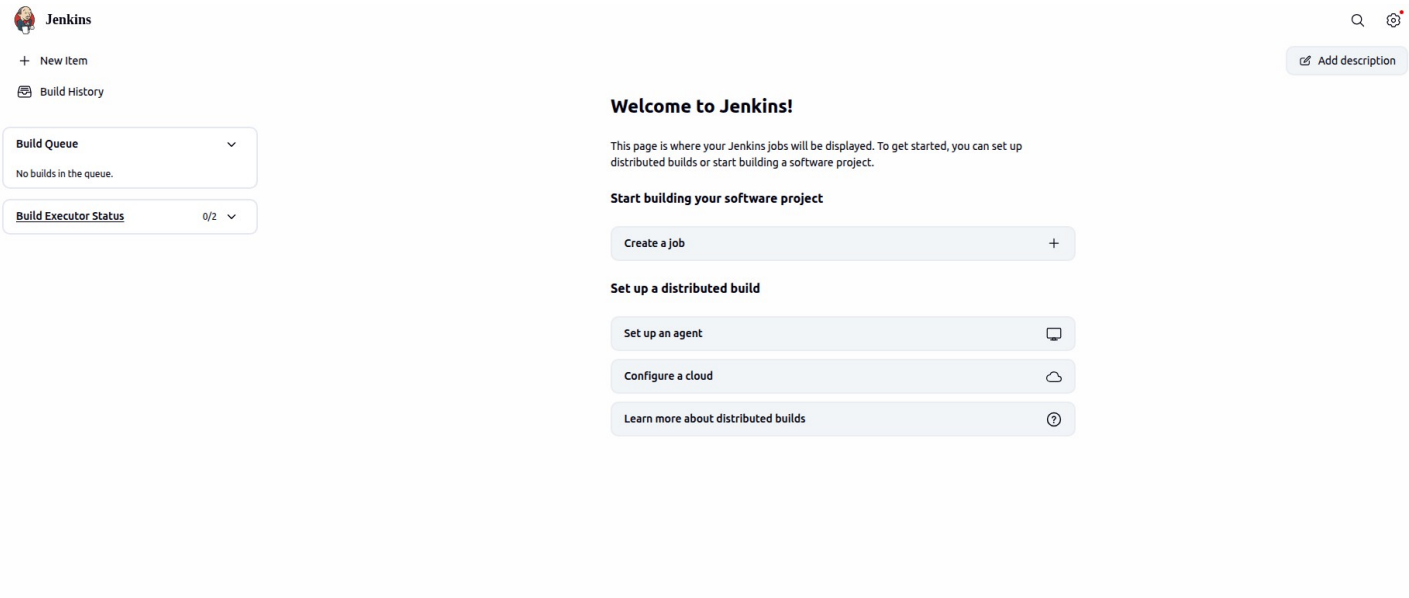
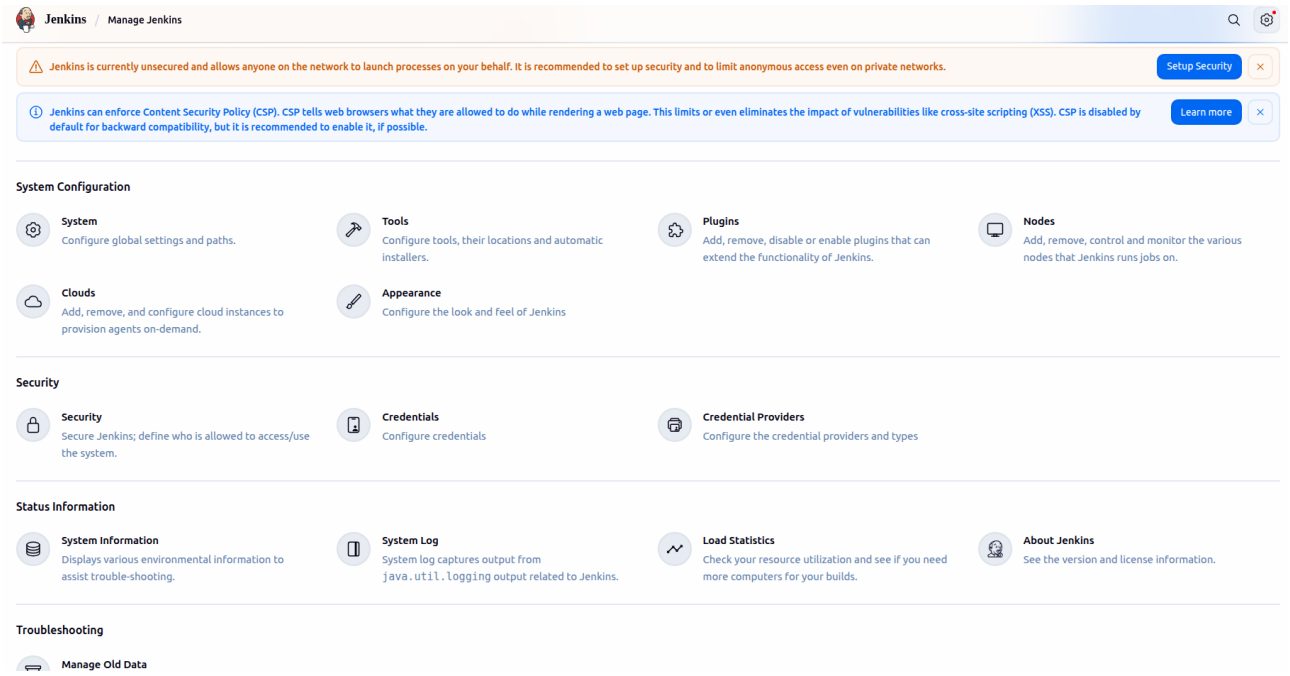
Program: Jenkins Setup and tomcat installation

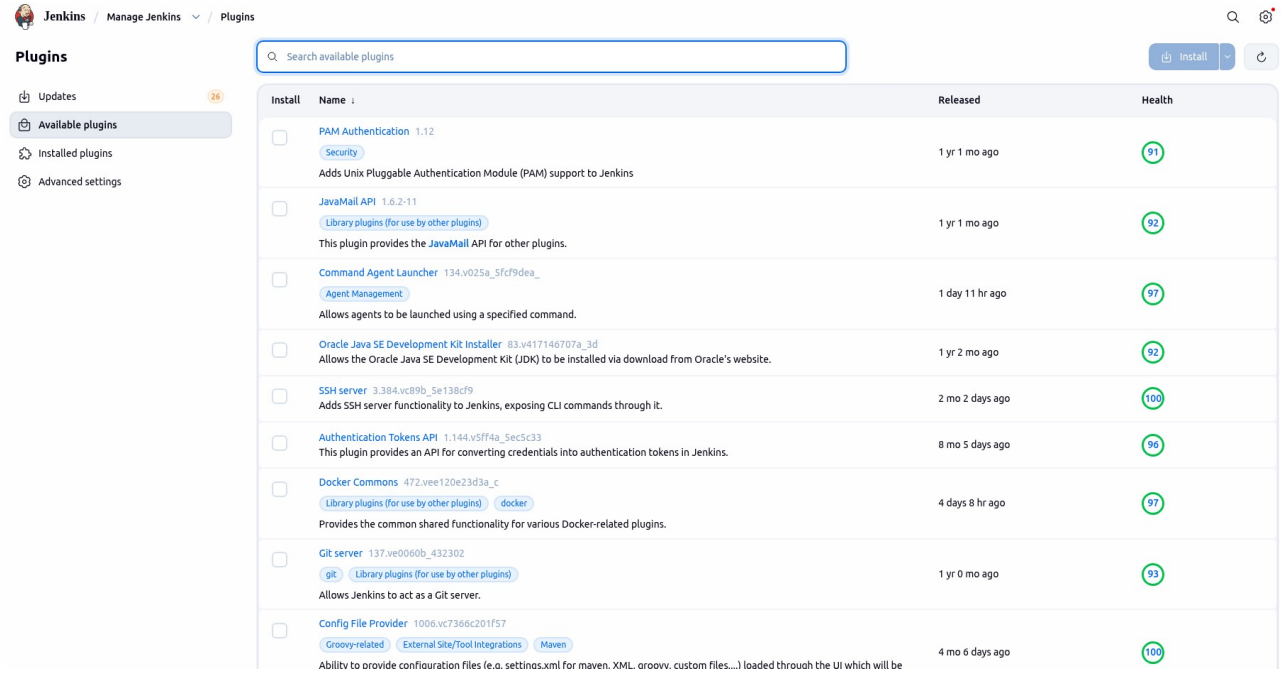
Step 1: Login to Jenkins Account



Step 2: Go to Manage Jenkins

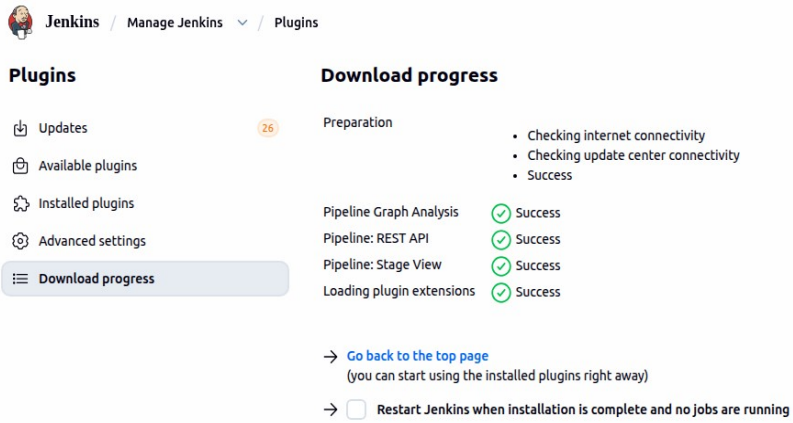


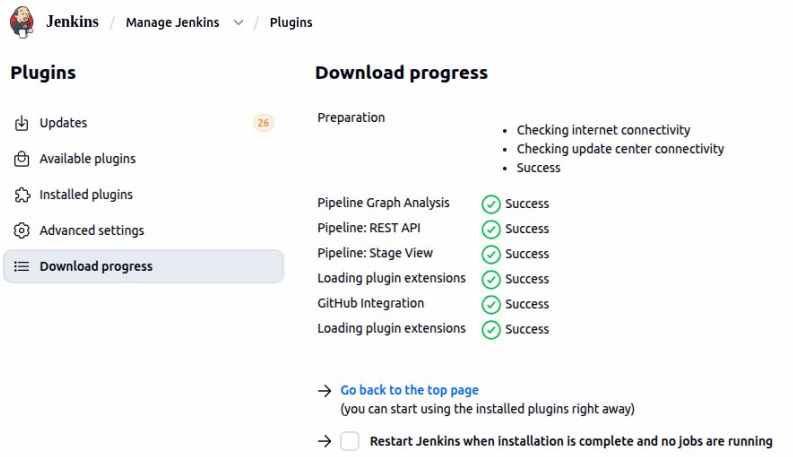
Step 3: Go to Plugins -> Available plugins

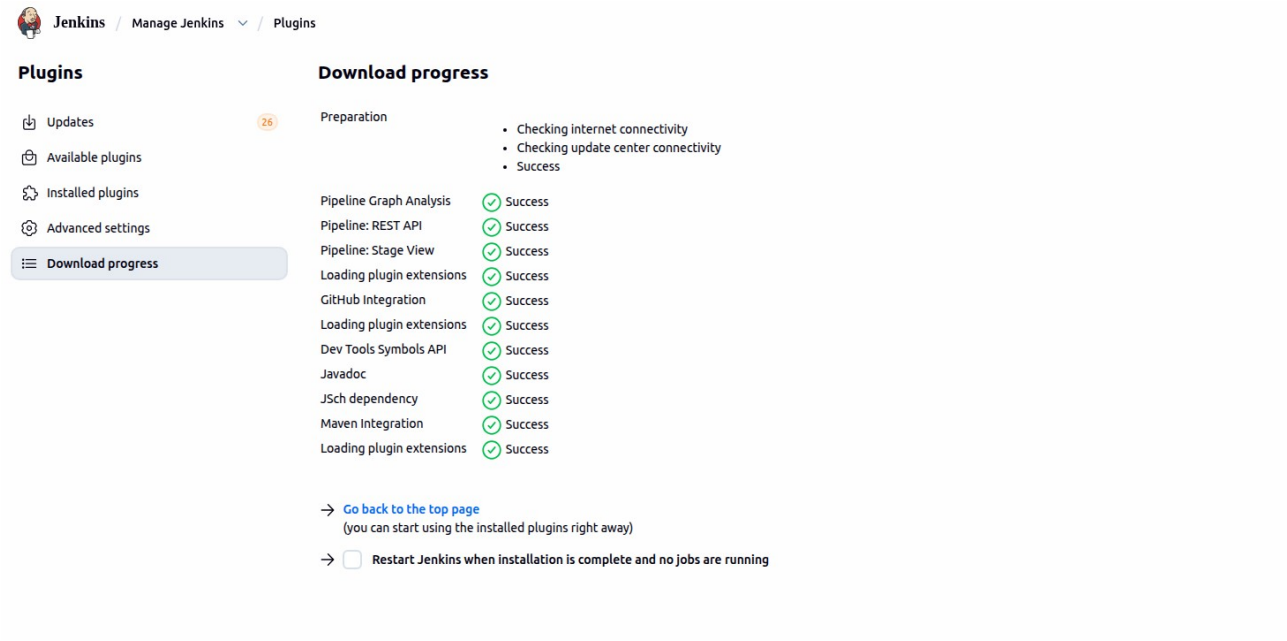


Step 4: Install the following plugins if not preset in available plugins

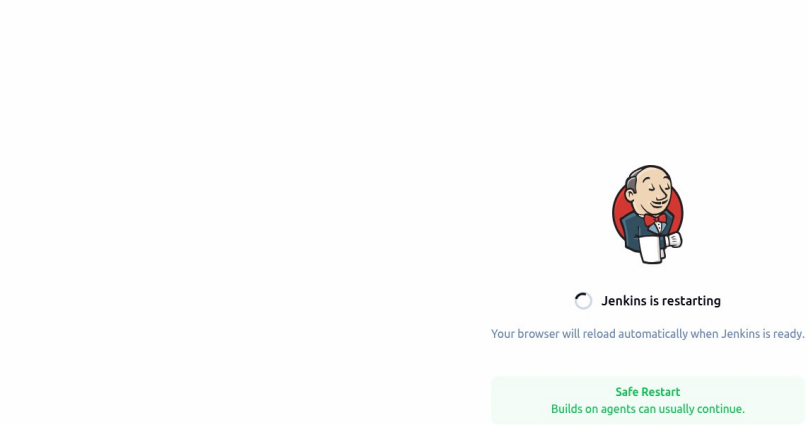
1.Pipline Stage View
2.Github Integration
3.Maven
4.Gradle







Step 5: Once installation is complete, restart jenkins using safeRestart



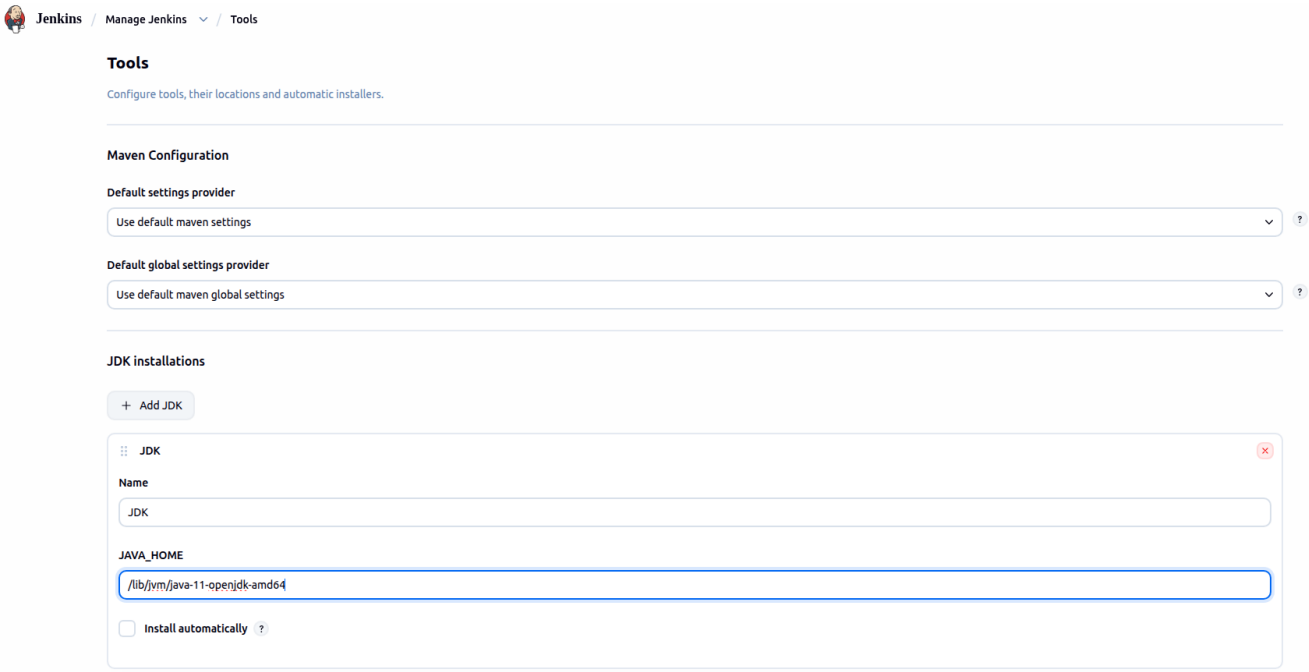
Step 6: Now configure tools: Manage Jenkins -> Tools

-Add JDK

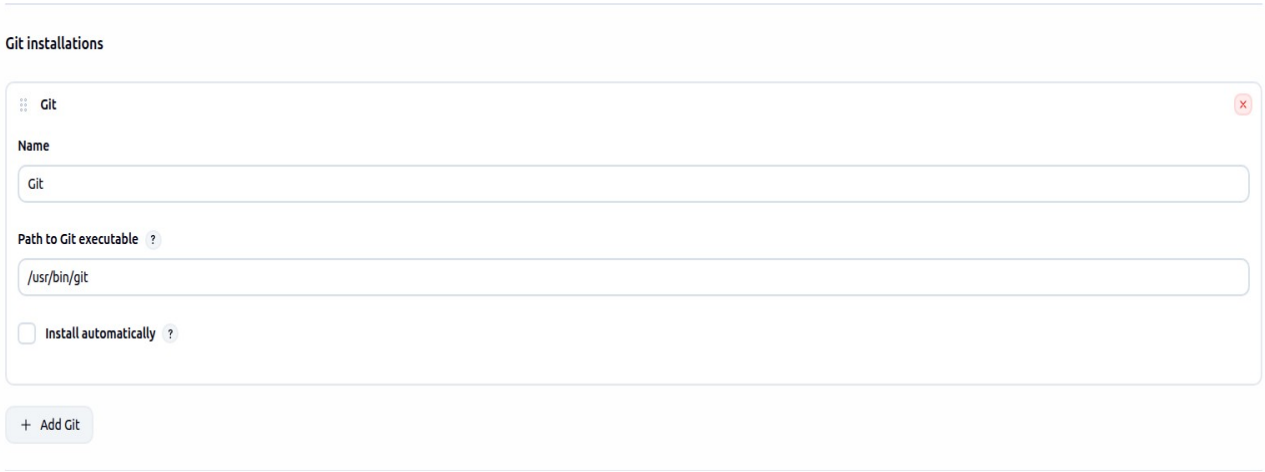
-Provide name and path.

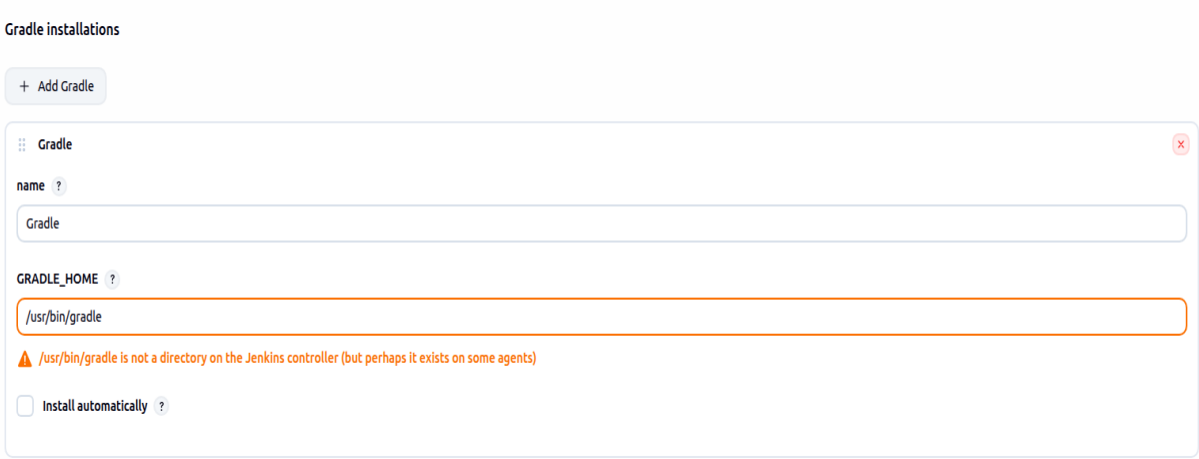
-The location can be found at:

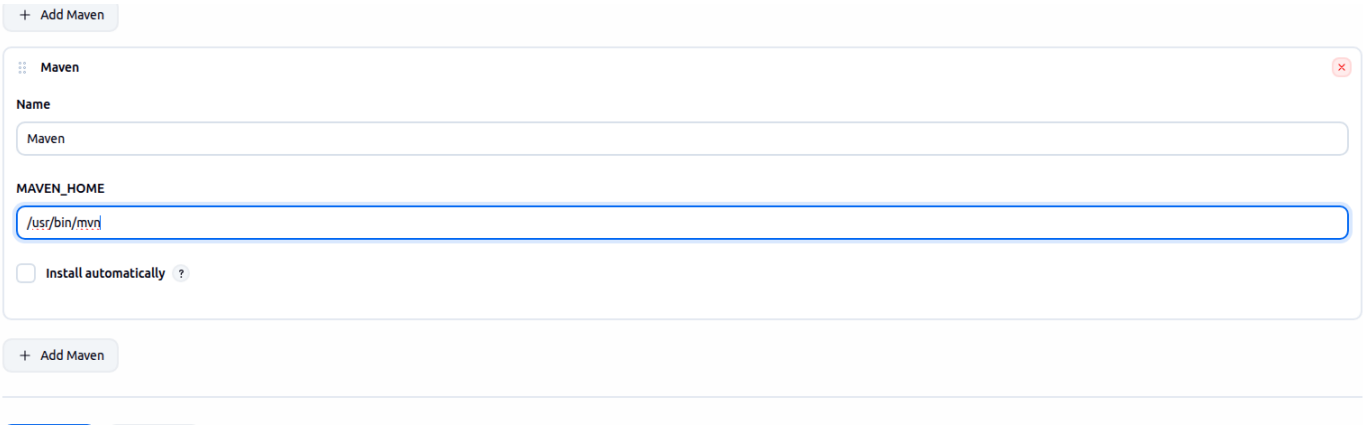
files -> other location -> computer -> lib -> jvm -> jdk



Step 7: Provide paths and names for the following tools:
-Git, Maven, Gradle
The paths for these can be determined using the cmnd which in terminal.

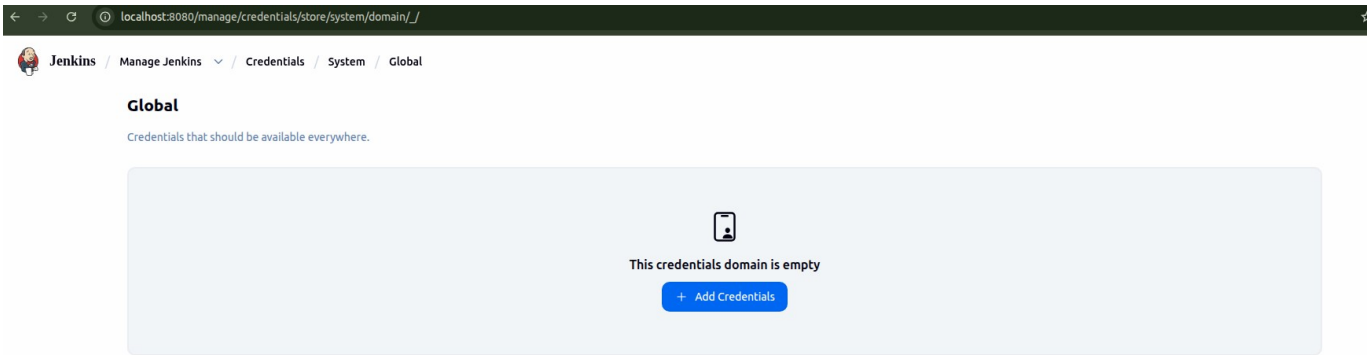




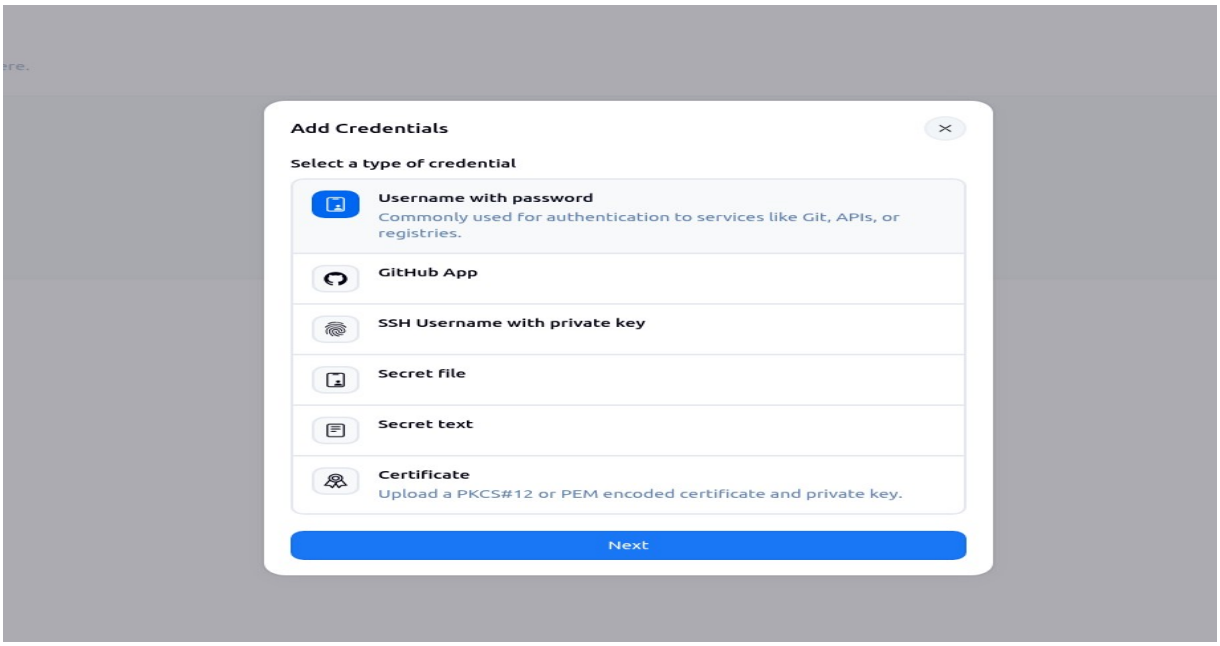


Step 8: Add credentials for github

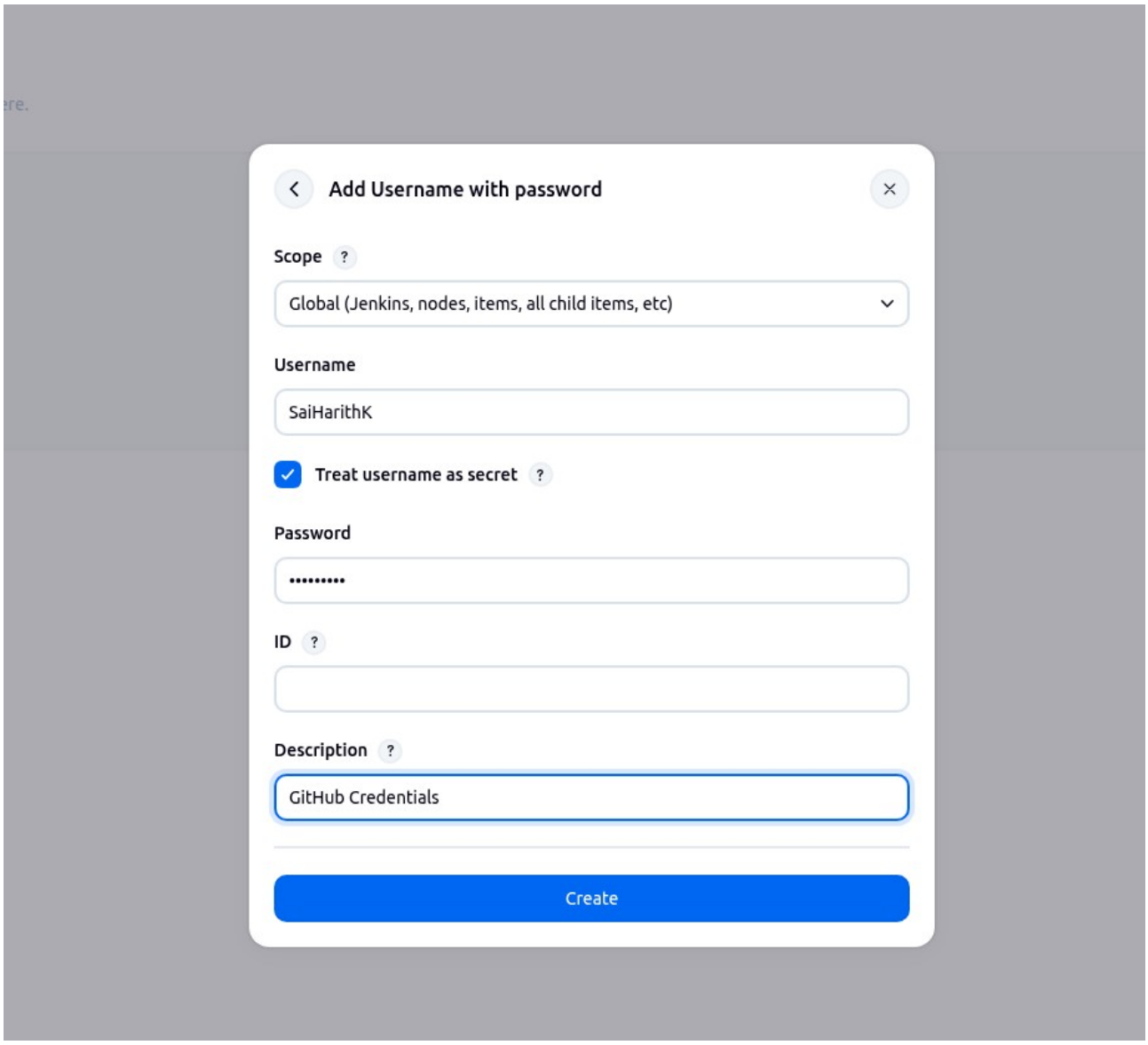
-Manage Jenkins -> credentials



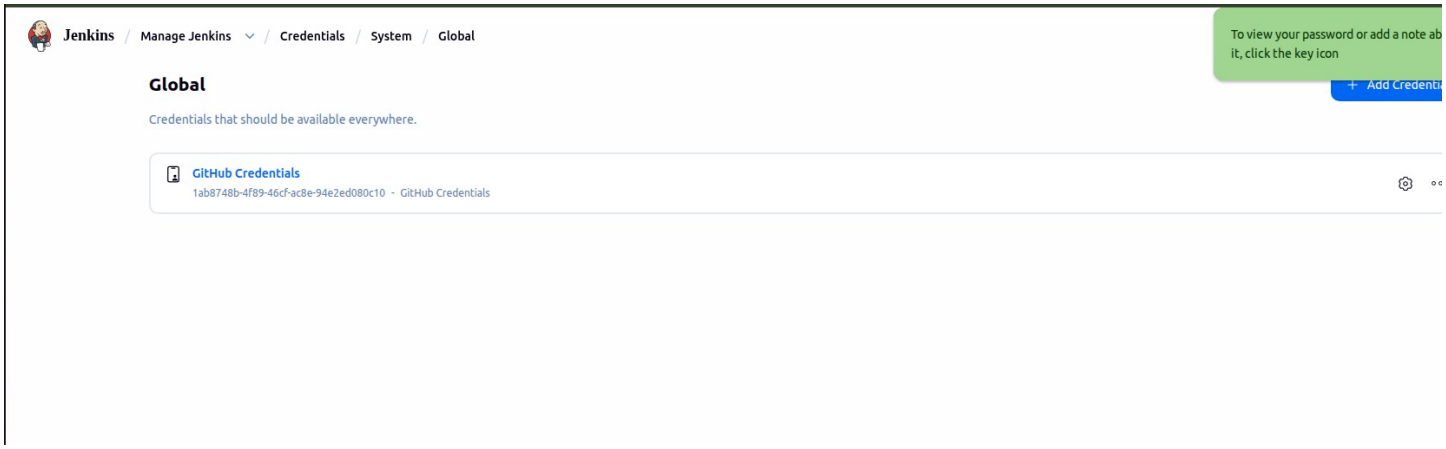
Select username with passwrod.



Provide github credentials and description.

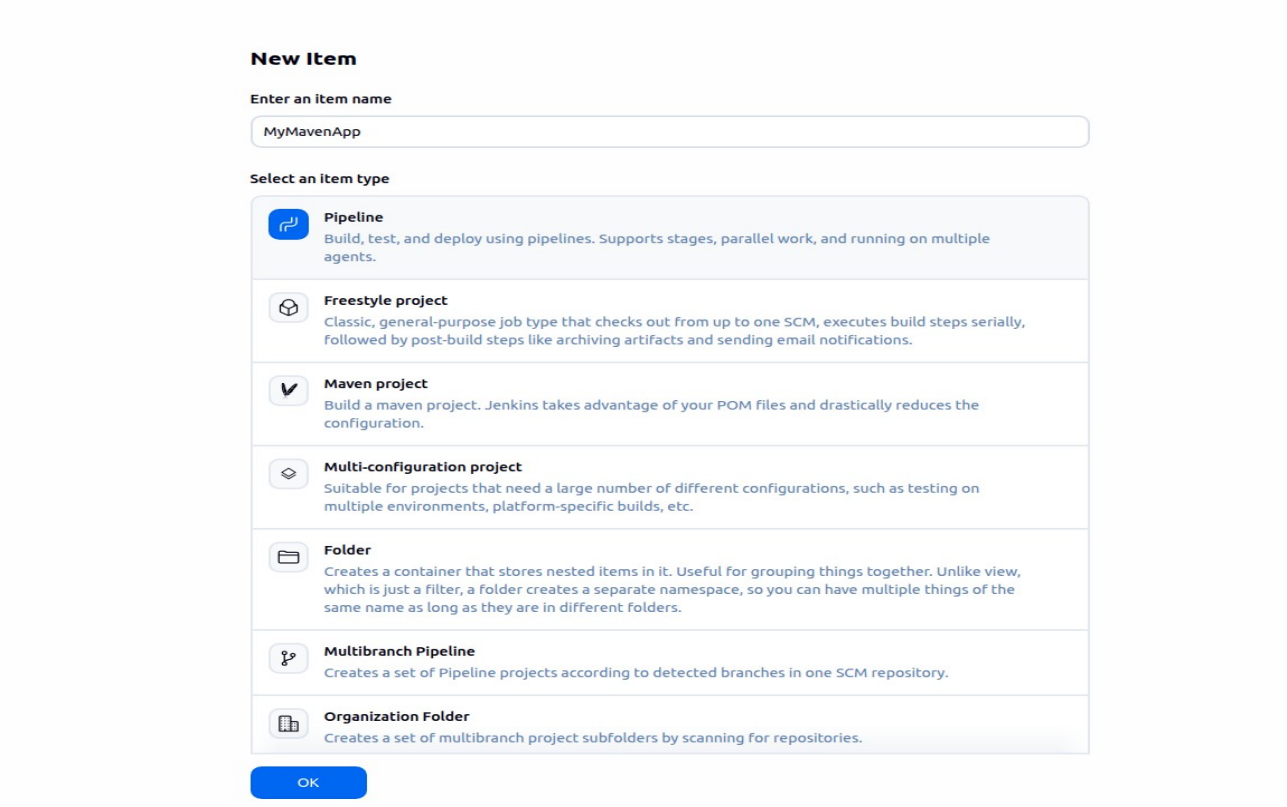


Click on apply and check whether it has been created succesfully.



Step 9: Create a new Pipeline

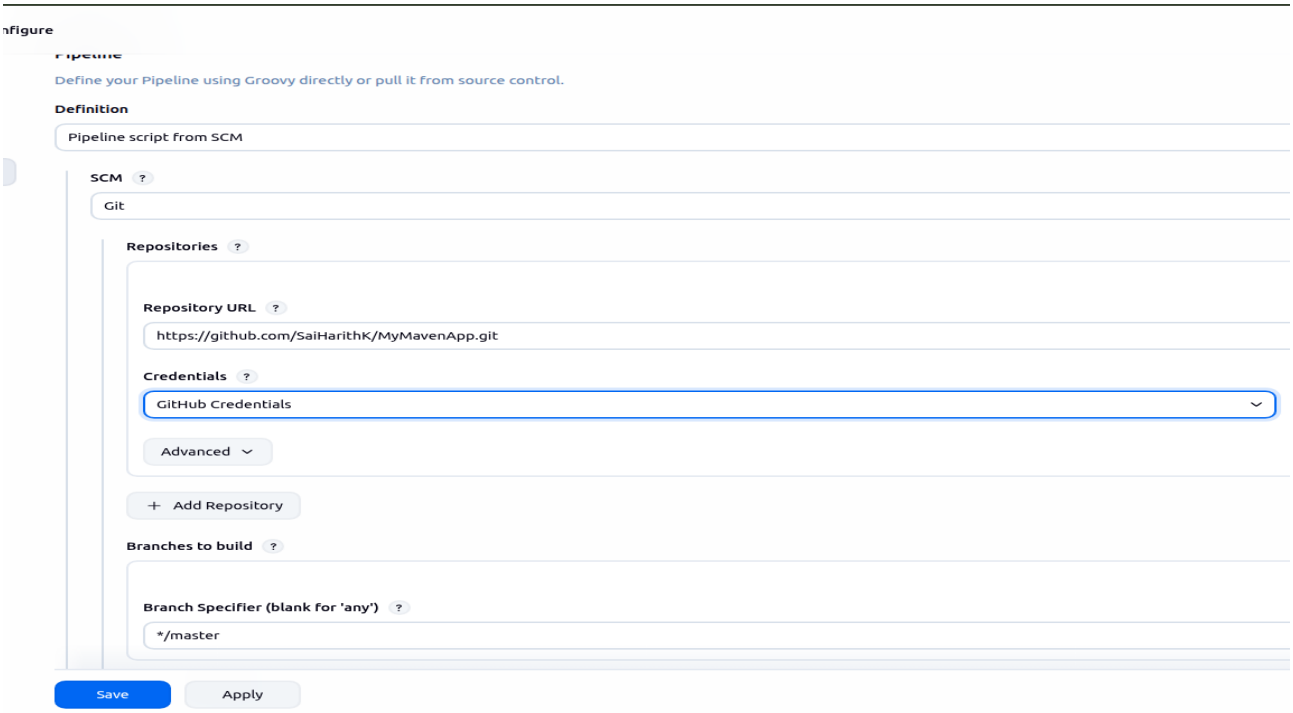
- Go to dashboard -> Add New Item



Provide name and select pipeline.

Step 10: Configure pipeline

-Select pipeline script from scm
-Select git for scm
-Select github credentials created previously
-Apply and save



Step 11: Create jenkins file
- Create it inside the poject, and add the following code:

Step 12: Edit pom.xml
- Make sure source and target versions are set to 11.
-Push the updated files to your github repo.

Step 13: Build Pipeline

Tomcat Installation:

Step 1: Open Terminal
- use sudo -i to swtich to root user
- sudo apt update && sudo apt upgrade -y -> updates all packages to latest versions without asking for permissions.

```
harith@harith-VirtualBox:~$ sudo -i
[sudo] password for harith:
root@harith-VirtualBox:~# sudo apt update && sudo upgrade -y
Hit:1 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:5 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
110 packages can be upgraded. Run 'apt list --upgradable' to see them.
sudo: upgrade: command not found
root@harith-VirtualBox:~# wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz
--2026-04-14 19:39:39--  https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12715996 (12M) [application/x-gzip]
Saving to: ‘apache-tomcat-9.0.95.tar.gz’

apache-tomcat-9.0.95.tar.gz                         100%[===================================================================================

2026-04-14 19:42:55 (63.6 KB/s) - ‘apache-tomcat-9.0.95.tar.gz’ saved [12715996/12715996]

root@harith-VirtualBox:~#
```

Step 2: Download Apache Tomcat

- Move intp tmp : cd tmp/ and enter the followng cmnd:
  “wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz “

```
root@harith-VirtualBox:~# wget https://archive.apache.org/dist/tomcat/tomcat-9/v
9.0.95/bin/apache-tomcat-9.0.95.tar.gz
--2026-04-15 15:23:29--  https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95
/bin/apache-tomcat-9.0.95.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a
:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... con
nected.
HTTP request sent, awaiting response... 200 OK
Length: 12715996 (12M) [application/x-gzip]
Saving to: ‘apache-tomcat-9.0.95.tar.gz’

      apache-tomcat   1%[                    ] 132.70K   121KB/s
```

Step 3: Create new tomcat directory
- cd ..
- cd /opt
- mkdir tomcat

Step 4: Extract the tomcat files
- “ sudo tar -xvzf apache-tomcat-9.0.95.tar.gz -C /opt/tomcat –strip-components=1”

Step 5: Make changes to server.xml
- cd /opt/tomcat/conf
- ls (check if server.xml is present)
- gedit server.xml
- Change port to 8090 from 8080, as Jenkins uses 8080.

```
        Define a non-SSL/TLS HTTP/1.1 Connector on port 8080
    -->
    <Connector port="8090" protocol="HTTP/1.1"
               connectionTimeout="20000"
               redirectPort="8443"
               maxParameterCount="1000"
               />
    <!-- A "Connector" using the shared thread pool-->
    <!--
    <Connector executor="tomcatThreadPool"
               port="8080" protocol="HTTP/1.1"
               connectionTimeout="20000"
               redirectPort="8443"
```

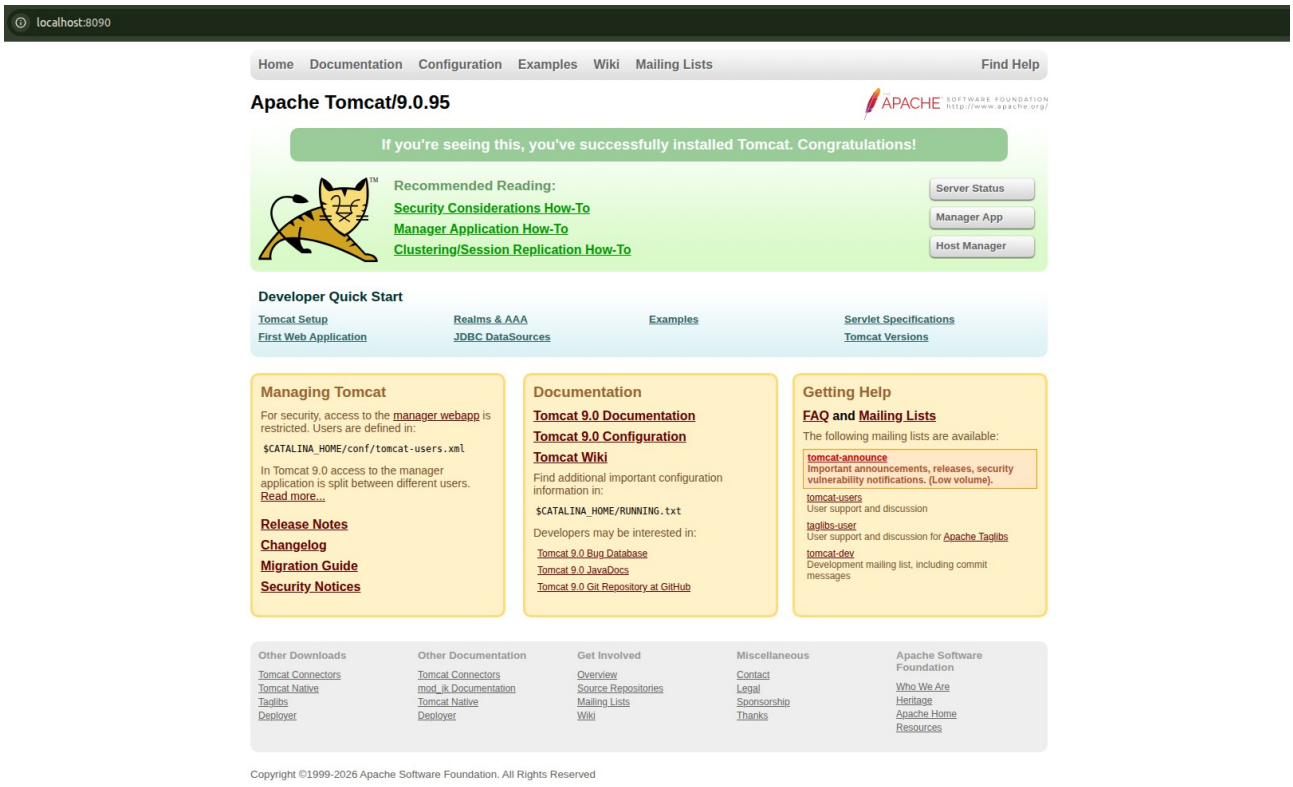
Step 6: Edit tomcat-users.xml to add user details:
- gedit tomcat-users.xml
- Add a username and password(user: admin, password: Admin)

```
-->
<tomcat-users xmlns="http://tomcat.apache.org/xml"
              xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
              xsi:schemaLocation="http://tomcat.apache.org/xml tomcat-users.xsd"
              version="1.0">
              <user username="admin" password="Admin" roles="manager-gui"/>
<!--
  By default, no user is included in the "manager-gui" role required
```

Step 7: Start Tomcat Server
- cd /opt/tomcat/bin
- ls
- sh startup.sh

```
root@harith-VirtualBox:/opt/tomcat/bin# sh startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli
.jar
Using CATALINA_OPTS:
Tomcat started.
root@harith-VirtualBox:/opt/tomcat/bin#
```

Step 8: Verify if Installation was successful
- Go to localhost:8090 on your browser
- Use Manager App to login. (Use login credentials added previously to tomcat-users.xml)



localhost:8090/manager/html

Tomcat Web Application Manager

Message:	OK

Manager

List Applications	HTML Manager Help	Manager Help	S

Applications

Path	Version	Display Name	Running	Sessions	Commands
/	*None specified*	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/MavenAnsibleWebApp1	*None specified*	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/MyMavenWebApp001	*None specified*	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	*None specified*	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	*None specified*	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	*None specified*	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/manager	*None specified*	Tomcat Manager Application	true	1	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

Deploy

Deploy directory or WAR file located on server

Context Path:

Version (for parallel deployment):

XML Configuration file path:

WAR or Directory path:

Deploy